

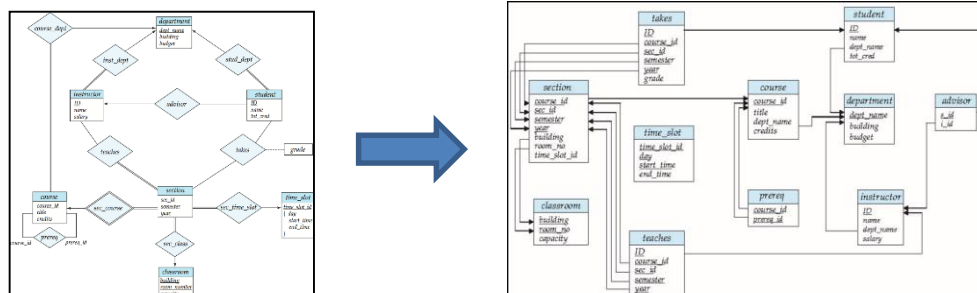
Database Design and Entity-Relationship Diagrams

Modelling real world systems with Entities and Relationships

Chapter 6 [7] in 7th [6th] edition of textbook

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These slides have been prepared by Anne Haxthausen, partly reusing/modifying slides by Flemming Schmidt. Some examples come from Silberschatz, Korth, Sudarshan, 2010.



Database Design

■ Three main phases

1. *Conceptual design:*

- Is the process of constructing a *conceptual model* of the data used in an enterprise, without taking any decisions on how the data should be logically or physically implemented in a DBMS.
- It is common to use *Entity-Relationship (E-R) models* as conceptual models. An ER model describes the entities and relationships that should be captured in the database, but without saying how these entities and their relationships should be captured in tables.

2. *Logical design:*

- Is the process of constructing a *logical model* of the data, for a specific type of databases (e.g. relational, object, or ...), but *independent of the DBMS* (e.g. MySQL, DB2, Oracle etc).
- For relational databases, the logical model is a *relational database schema*, i.e. a collection of relation/table schemas.

3. *Physical design:*

- Is the process of describing how the database *should* be implemented on a *specific DBMS*. The physical features of the database system are specified. These include e.g. the choice of file organization and indexes.

■ Objectives today:

1. Show how to use *Entity-Relationship (E-R) modelling* to build a conceptual data model.
2. Show how *to convert* an E-R model into a logical design for the relational model.

E-R Modelling with E-R Diagrams

- Entity Sets and Relationship Sets
- Attributes
- Order, Cardinality and Participation
- Keys for Entities and Relationships
- Entity-Relationship (E-R) Diagrams
- Strong and Weak Entities
- E-R Diagram for the University Database
- Summary of the Textbook Adapted UML Notation
- Other Diagram Notations

Entities and Relationships

The data of an enterprise can be modelled as entities and relationships.

- **An entity e**
 - Is a thing/object, e.g. a specific student.
- **An entity set E**
 - Is a set of entities representing objects of the same type (having the same attributes).
 - Examples: the set *student* of all students.
- **Entity attributes**
 - Each entity has properties called **entity attributes**.
 - Examples of *student* Attributes: *ID* and *name*.
- **A relationship**
 - Is an association $(e_1, \dots, e_n)$ among several entities e_i .
- **A relationship set among entity sets $E_1, \dots, E_n$**
 - Is a set of relationships of the same type, i.e. a subset $\{(e_1, \dots, e_n) \mid e_i \in E_i\}$.
 - Examples: *advisor*
- **Relationship attributes**
 - Each relationship can also have properties called **relationship attributes**.
 - Example of an *advisor* attribute: *date* (for start date).

Two Entity Sets

ID	name
76766	Crick
45565	Katz
10101	Srinivasan
98345	Kim
76543	Singh
22222	Einstein

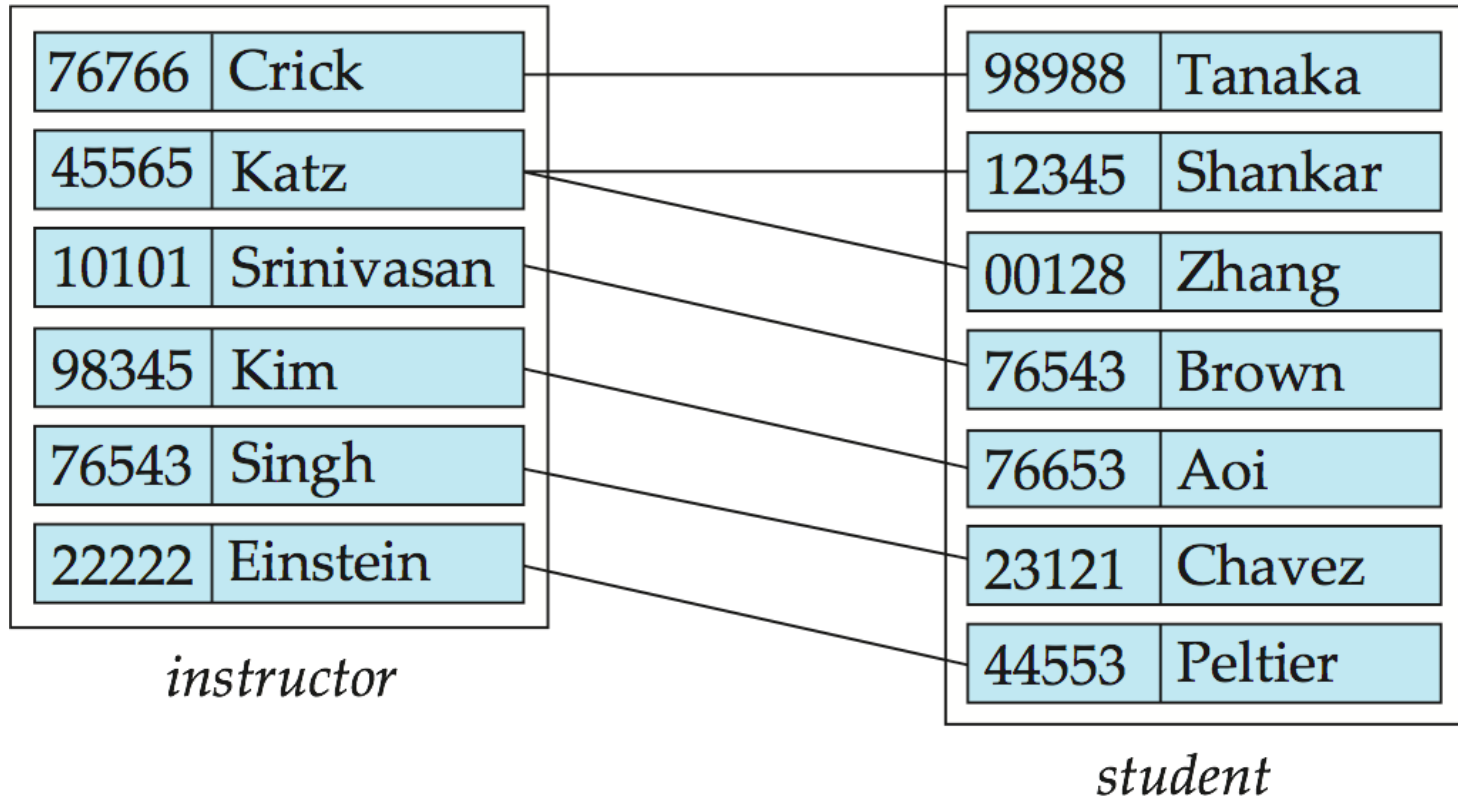
instructor

ID	name
98988	Tanaka
12345	Shankar
00128	Zhang
76543	Brown
76653	Aoi
23121	Chavez
44553	Peltier

student

PS. In this presentation names in the University Database have been altered slightly to conform with the textbook, e.g. ID instead of InstID and name instead InstName.

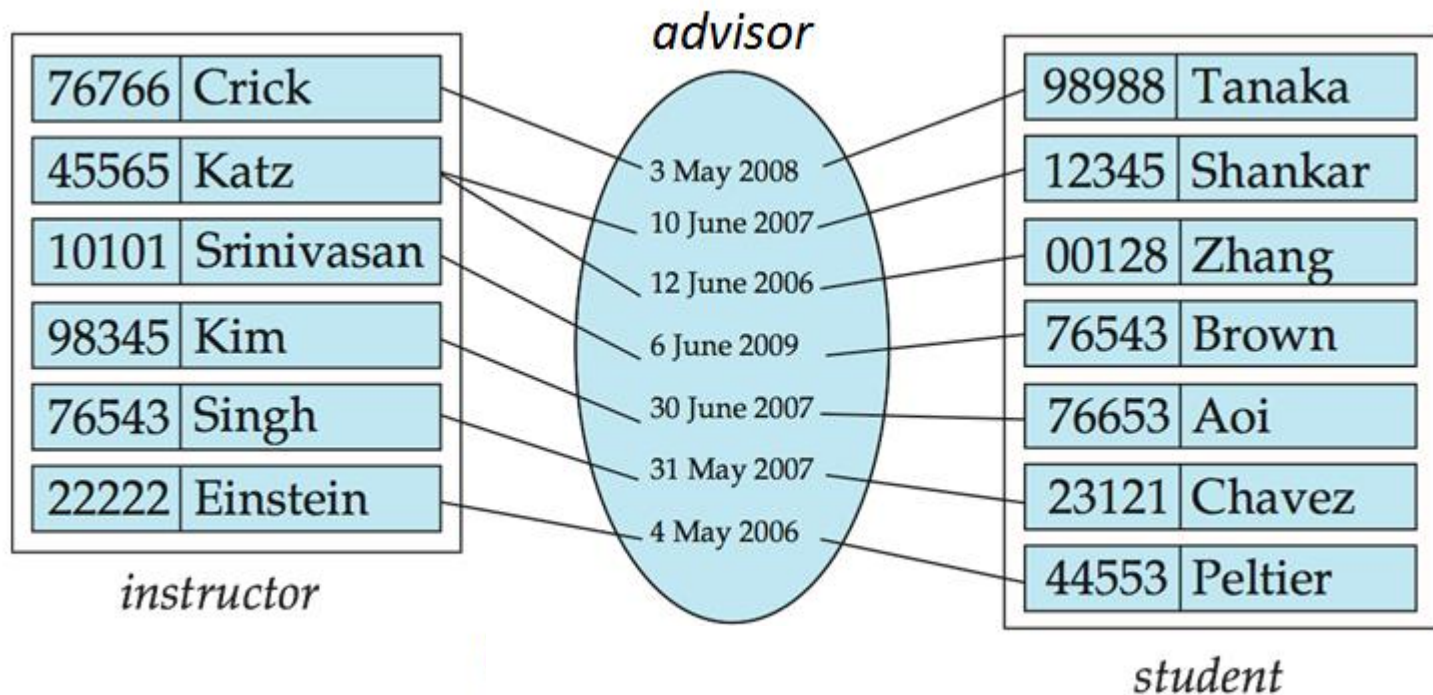
Relationship Set *advisor*



Each line illustrates a relationship.

Relationship Attributes - Example

- An *advisor* relationship between an *instructor* entity and a *student* entity may have the *attribute date*, which tracks, when the relationship was initiated.
- E.g., 4 May 2006 the instructor Einstein became advisor for the student Peltier.



Attribute Types

■ Attributes

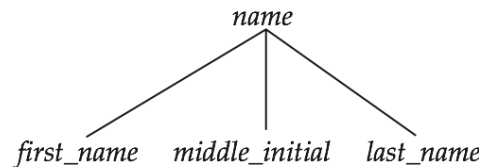
- are descriptive properties possessed by all elements of an entity set or a relationship set.

■ The Domain of an Attribute

- is the set of permitted values for the attribute.

■ Attribute Types

- *Simple (atomic)* attributes.
- *Composite* attributes, which consist of component attributes.

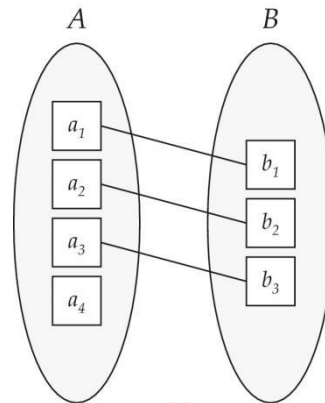


- *Multivalued* attributes, which is a set of attribute values. E.g. *phone_numbers*.
- *Derived* attributes, which are computed from other attributes.

Order, Cardinality and Participation

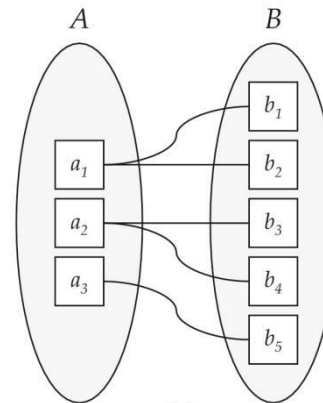
- Order/Degree for relationship set:
 - Is the number of entities (entity sets) being associated in a relationship (set)
 - If two entities (entity sets) are associated then the relationship (set) is *Binary*
 - If three entities (entity sets) are associated then the relationship (set) is *Ternary* etc.
- Cardinality for binary relationship set between entity sets E_1 and E_2 :
 - **one-to-one**: An element in E_1 can be associated with at most **one** (0..1) element in E_2 , and vice versa.
 - **one-to-many**: An element in E_1 can be associated with **many** (0..*) elements in E_2 , and an element in E_2 can be associated with at most **one** (0..1) element in E_1 .
 - **many-to-one**: An element in E_1 can be associated with at most **one** (0..1) element in E_2 , and an element in E_2 can be associated with **many** (0..*) elements in E_1 .
 - **many-to-many**: An element in E_1 can be associated with **many** (0..*) elements in E_2 , and an element in E_2 can be associated with **many** (0..*) elements in E_1 .
- Participation in binary relationship set between entity sets E_1 and E_2 :
 - **Total Participation** of E_1 in a relationship set between E_1 and E_2 :
All elements of E_1 participate in at least one association with elements in E_2 .
(So associations with 0 elements is not allowed.)
 - **Partial Participation** of E_1 in a relationship set between E_1 and E_2 :
Some elements in E_1 may not participate in any association with elements in E_2 .
(So association with 0 elements is allowed.)

Cardinality and Participation



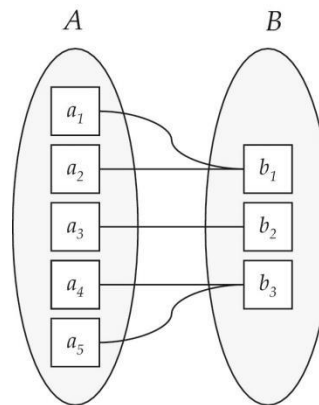
(a)

one-to-one
A partial, B total



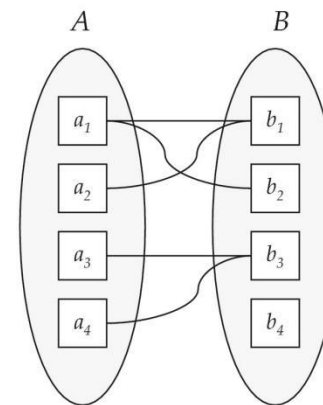
(b)

one-to-many
A total, B total



(a)

many-to-one
A total B total



(b)

many-to-many
A total, B partial

Key for an Entity Set

- A (Super) Key of an Entity Set
 - A set of one or more attributes whose values uniquely determine each Entity element in an Entity Set.
- A Candidate Key of an Entity Set
 - Is a Super Key which is minimal (i.e. has no other super key as a proper subset)
 - Example: *ID* is candidate key of *instructor*
- A Candidate Key is selected as Primary Key
 - Although several candidate keys may exist, one of the candidate keys is selected by to be the Primary Key.

Key for a Relationship Set

■ A Super Key of a Relationship Set

- The combination of primary keys of the participating entity sets **and the attributes of the relationship set** forms a super key of a relationship set.
- Example: *(student.ID, instructor.ID)* is a super key of *advisor*.

■ Deciding Candidate Keys of a Relationship Set

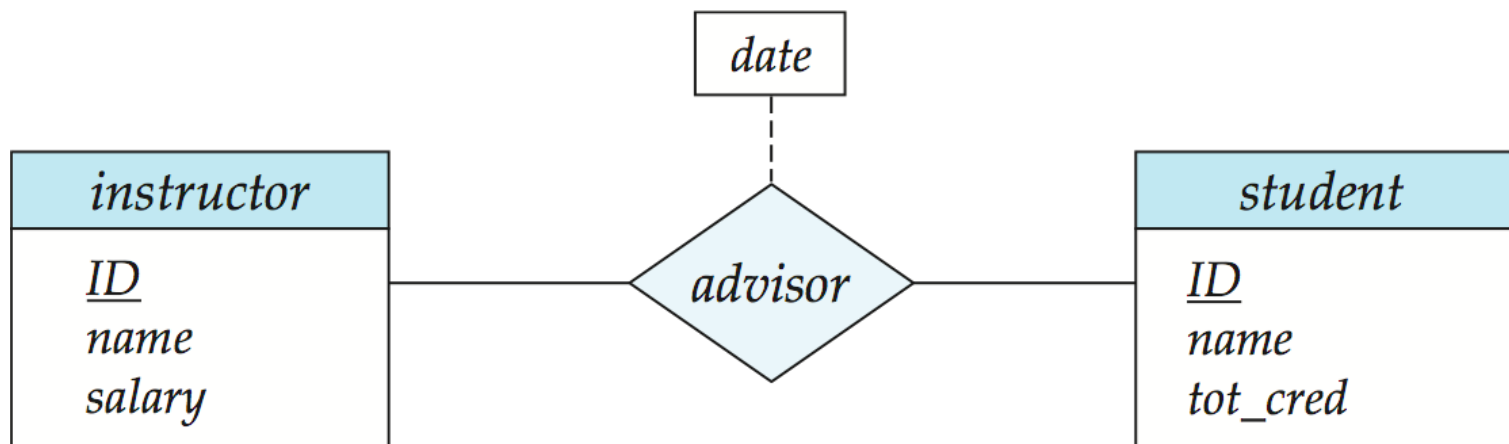
- Must consider the semantics of the relationship set to decide whether attributes of the relationship set should contribute to candidate keys. (Usually they do not contribute, but if it is multivalued it contributes, see demo exercise 6.4.)
- Must consider the cardinality of the relationship set when deciding what are the candidate keys. For instance, for *advisor*
 - *(student.ID, instructor.ID)* is a candidate key, if the *advisor* relation is many-many.
 - *student.ID* is a candidate key, if the *advisor* relation is many-one (with *student* on the many side).

■ Selecting the Primary Key

- Remember to consider the semantics of the relationship set when selecting the primary key, since there can be more than one candidate key.

Entity-Relationship Diagrams

- An E-R diagram is a diagrammatic representation of an E-R model. There are many *different* E-R diagram notations, such the Textbook's **Adapted UML Class Diagram** notation.
- *Entity sets* are represented by rectangles with a name in the blue top.
 - *Entity attributes* are listed inside the entity rectangle.
 - *Primary keys* are underlined.
- *Relationship sets* are represented by a diamond with a name listed inside, and lines connecting to the participating entities.
 - *Relationship attributes* are listed in a box, linked with a dashed line.

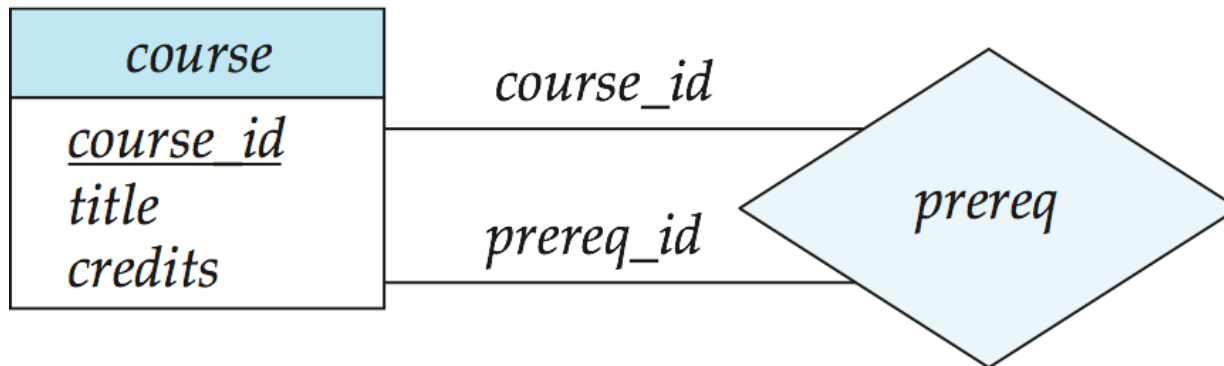


Notation to Express Complex Attributes

<i>instructor</i>	
<u><i>ID</i></u>	atomic attribute
<i>name</i>	composite attribute
<i>first_name</i>	
<i>middle_initial</i>	
<i>last_name</i>	
<i>address</i>	composite attribute
<i>street</i>	
<i>street_number</i>	
<i>street_name</i>	
<i>apt_number</i>	
<i>city</i>	
<i>state</i>	
<i>zip</i>	
{ <i>phone_number</i> }	multivalued attribute
<i>date_of_birth</i>	atomic attribute
<i>age</i> ()	derived attribute

Roles of Relationship Sets

- The entity sets of a relationship set need not be distinct
 - Each occurrence of an entity set plays a “role” in the relationship set.
 - The labels “*course_id*” and “*prereq_id*” are called *roles*.



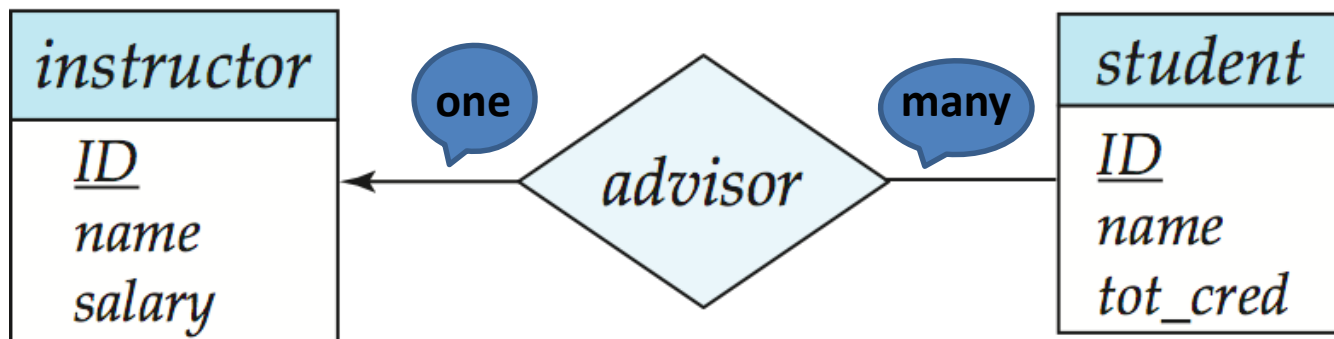
Drawing Cardinality Constraints

■ Drawing Cardinality Constraints

- We express cardinality constraints by drawing either a directed line ($\rightarrow$), signifying “one,” or an undirected line ($—$), signifying “many,” between the relationship set and the entity set.

■ Example:

- **One-to-Many** Relationship Set between *instructor* and *student*:
An instructor is associated with several students via *advisor*.
A student is associated with at most one instructor via *advisor*.



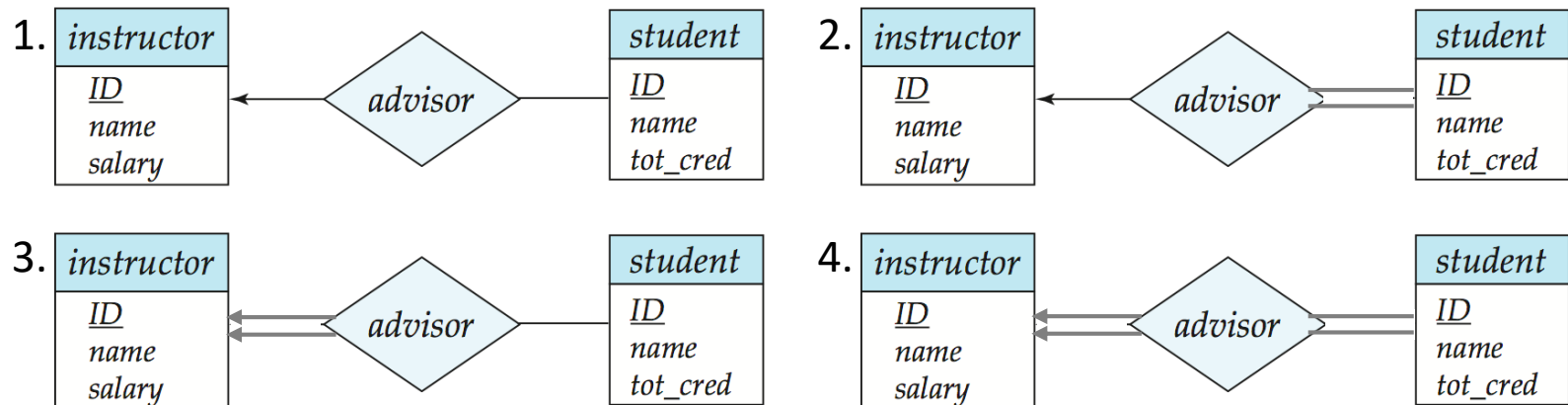
Participation of an Entity Set in a Relationship Set

■ Total Participation

- Every entity element participates in at least one relationship. This is shown by a **double line**.

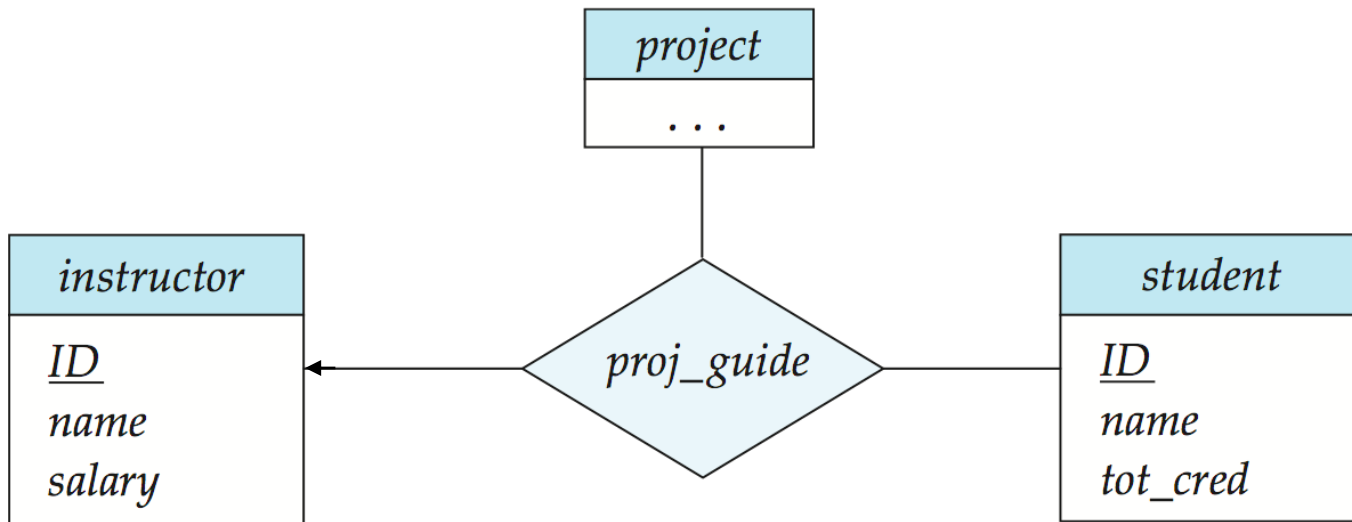
■ Partial Participation

- Some entity elements may not participate in any relationship. This is shown by a **single line**.



1. An instructor might not be advisor; A student might not have an advisor.
2. An instructor might not be advisor; Every student must have an advisor.
3. Every instructor must be advisor; A student might not have an advisor.
4. Every instructor must be advisor; Every student must have an advisor.

E-R Diagram with a Ternary Relationship Set

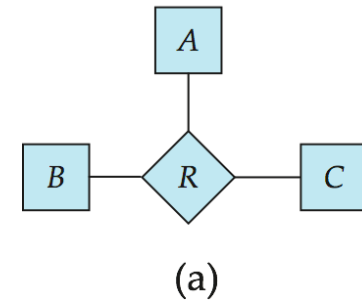


- A **ternary relationship (set)** is a relationship (set) between three entities (entity sets).
- We allow **at most one arrow** out of a ternary (or greater degree) relationship set. E.g. an arrow from *proj_guide* to *instructor* indicates each student has at most one guide for a given project.
- If there is more than one arrow, then there are several ways of defining the meaning. To avoid confusion at most one arrow is allowed.
- To avoid handling the complexities of a ternary relationship set it can be substituted by three binary relationship sets!

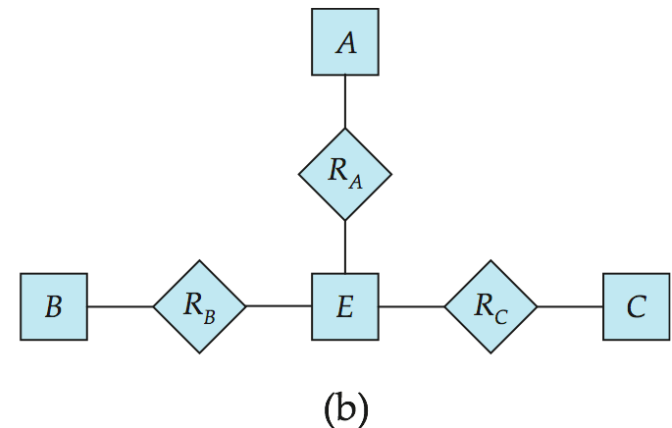
Convert Non-Binary Relationships to Binary

- In general, any non-binary relationship can be represented using binary relationships by creating an artificial entity set.

- Replace R between the entity sets A , B and C by an entity set E and three relationship sets:
 1. R_A , relating E and A
 2. R_B , relating E and B
 3. R_C , relating E and C
- Create a special identifying attribute for E
- Add any attributes of R to E
- For each relationship (a_i, b_i, c_i) in R , create:
 1. A new entity e_i in the entity set E
 2. Add (e_i, a_i) to R_A
 3. Add (e_i, b_i) to R_B
 4. Add (e_i, c_i) to R_C

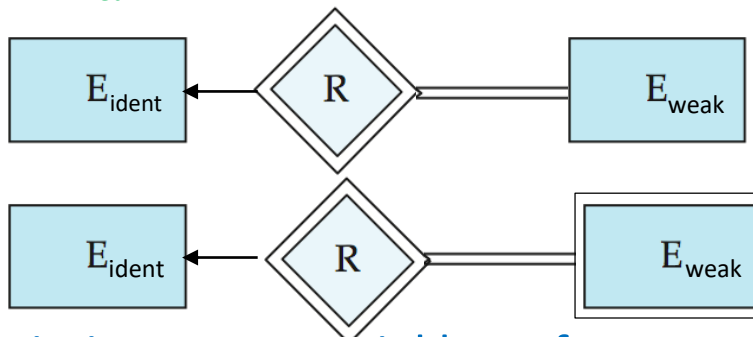


- Change the E-R Diagram accordingly



Strong and Weak Entity Sets

- A strong entity set
 - is one that can form its primary key from its own attributes.
- A weak entity set
 - is one that cannot form its primary key from its own attributes alone.
- Identifying entity set
 - The existence of a weak entity set E_{weak} depends on the existence of an **identifying entity set** E_{ident} . There must be an **identifying relationship set** R between E_{ident} and E_{weak} : a one-to-many relationship set R from E_{ident} to E_{weak} , with total participation of E_{weak} . Identifying relationship sets R are shown in double diamonds:



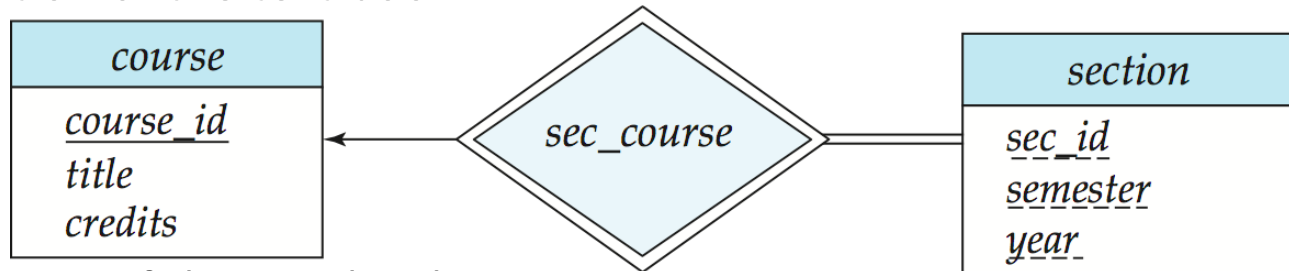
In 6th edition of book: single rectangle around E_{weak} .

In 7th edition of book: double rectangle around E_{weak} .

- Discriminator or partial key of E_{weak}
 - is a (sub)set of its attributes that can be used to distinguish an entity that depends on a particular strong entity.
- Primary key of E_{weak}
 - is formed by the primary key of E_{ident} plus the partial key of E_{weak} .

Weak Entity Sets - Example

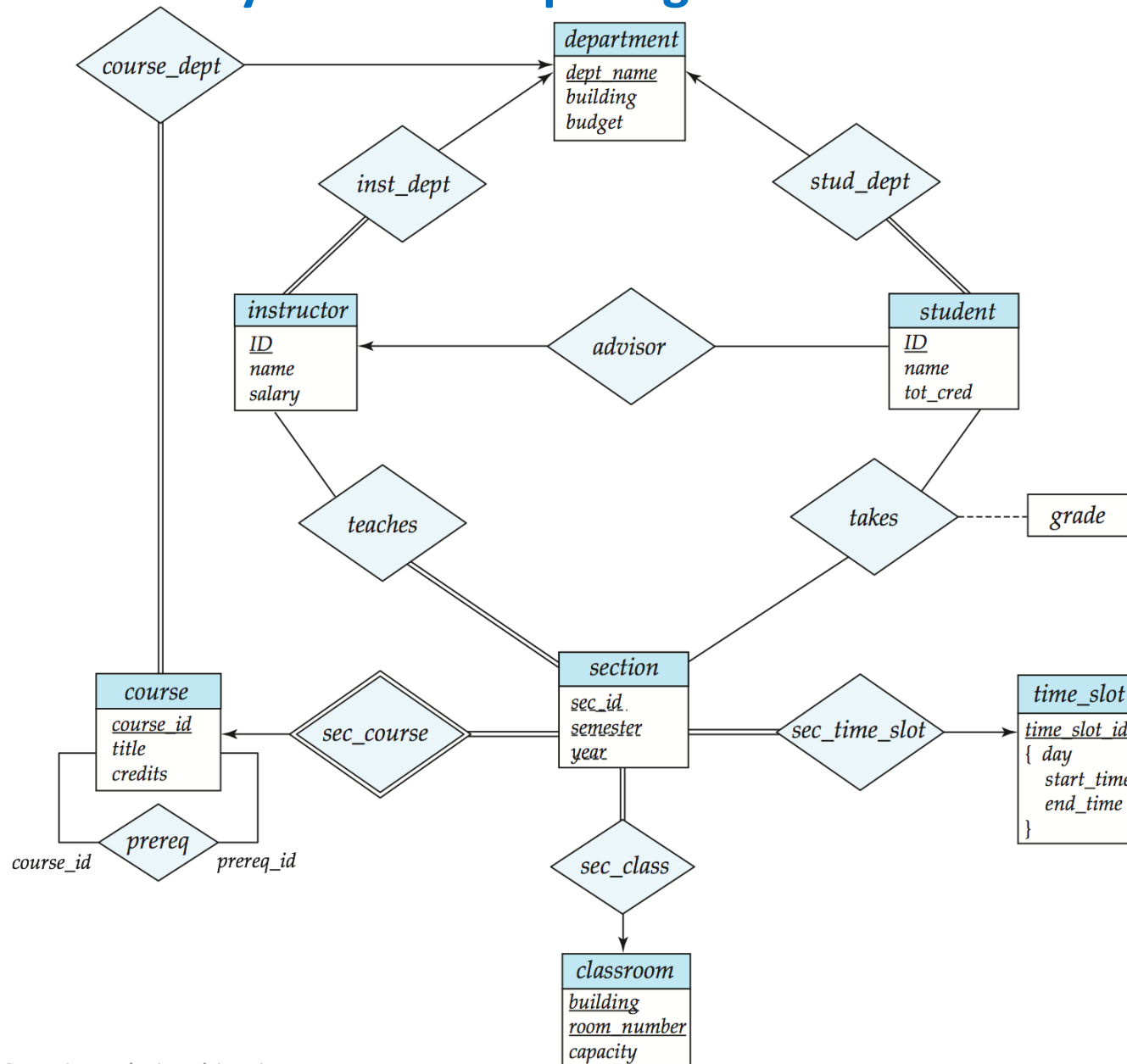
- *section* is a **weak entity set** having *course* as its **identifying entity set** and *sec_course* as its **identifying relationship set**.
- The **partial key** of a weak entity (set) is underlined with a dashed line.
- The **identifying relationship set** of a weak entity set is encapsulated in a double diamond.
- **Primary key** for section consists of: course id, sec id, semester, year
- In 6th edition of the text book:



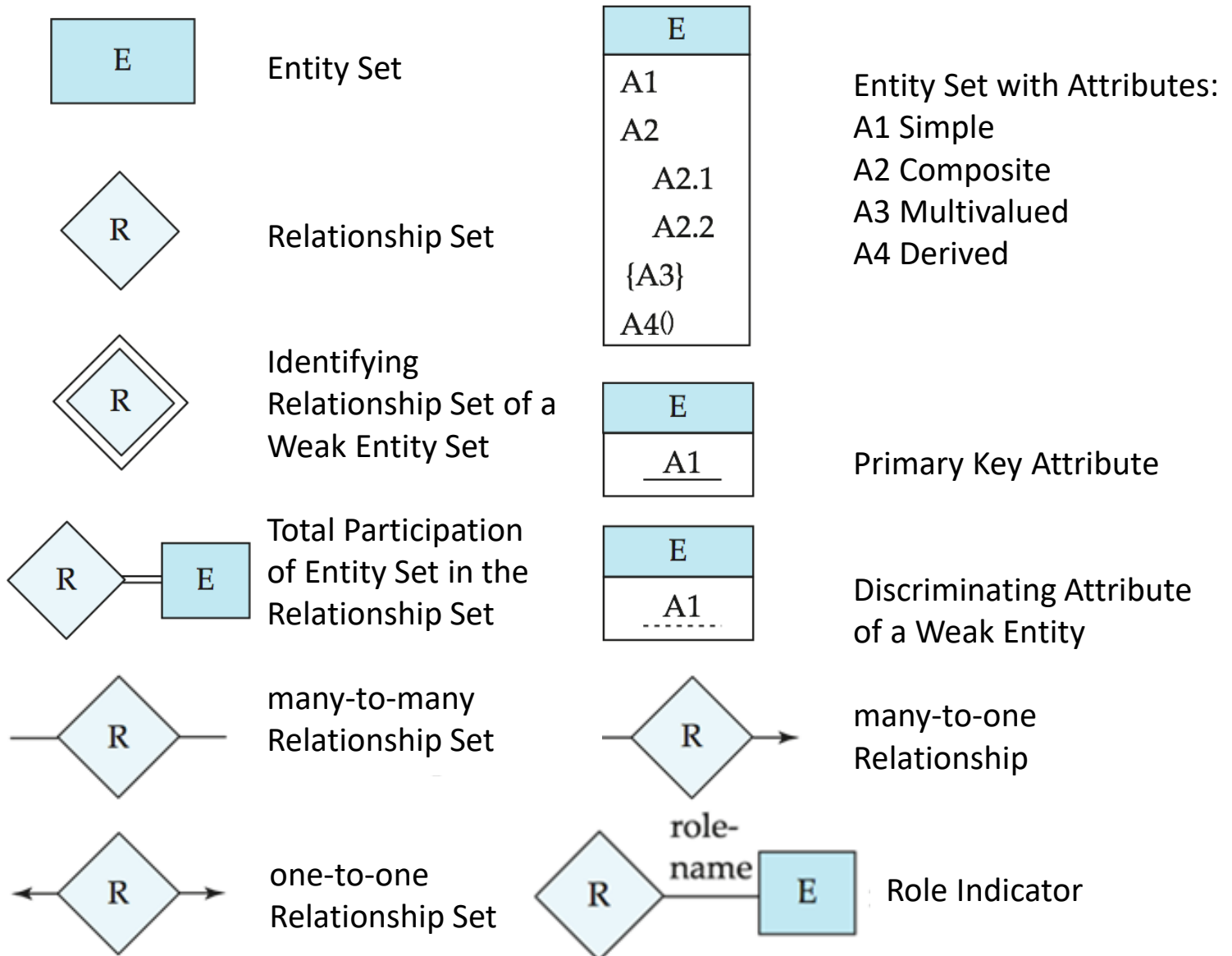
- In 7th edition of the text book:



Entity-Relationship Diagram for the University Database

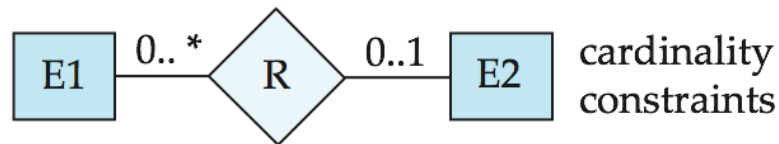
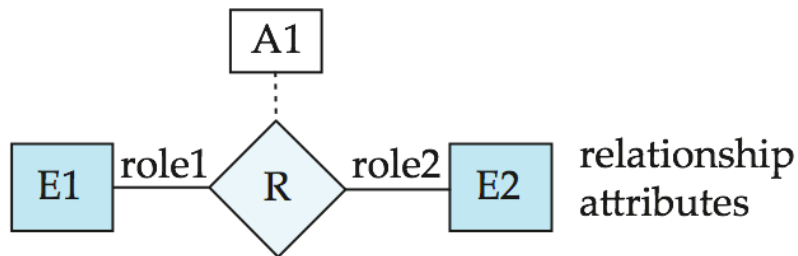
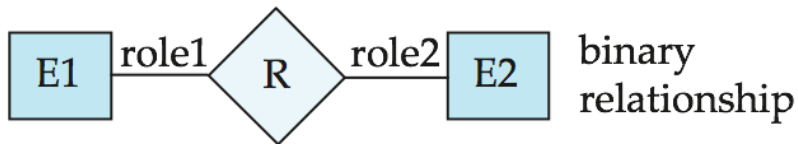
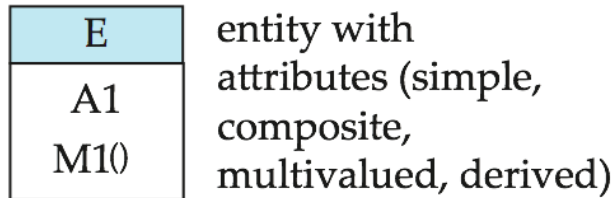


Summary of Textbook Adapted UML Notation

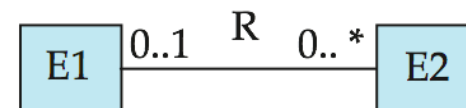
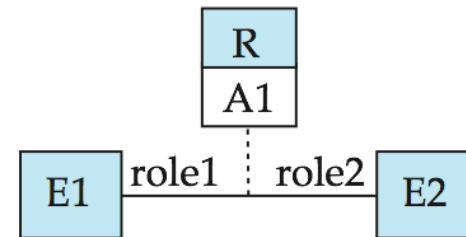
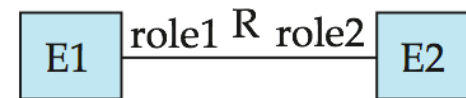
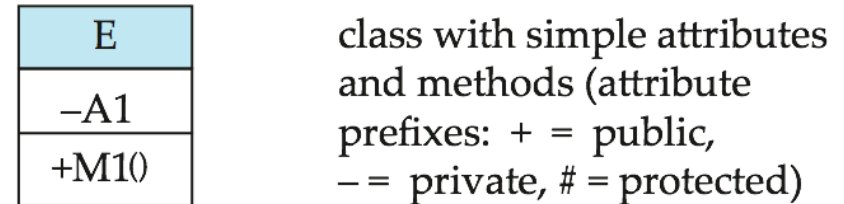


Other Diagram Notations: ER vs. UML Class Diagrams

ER Diagram Notation

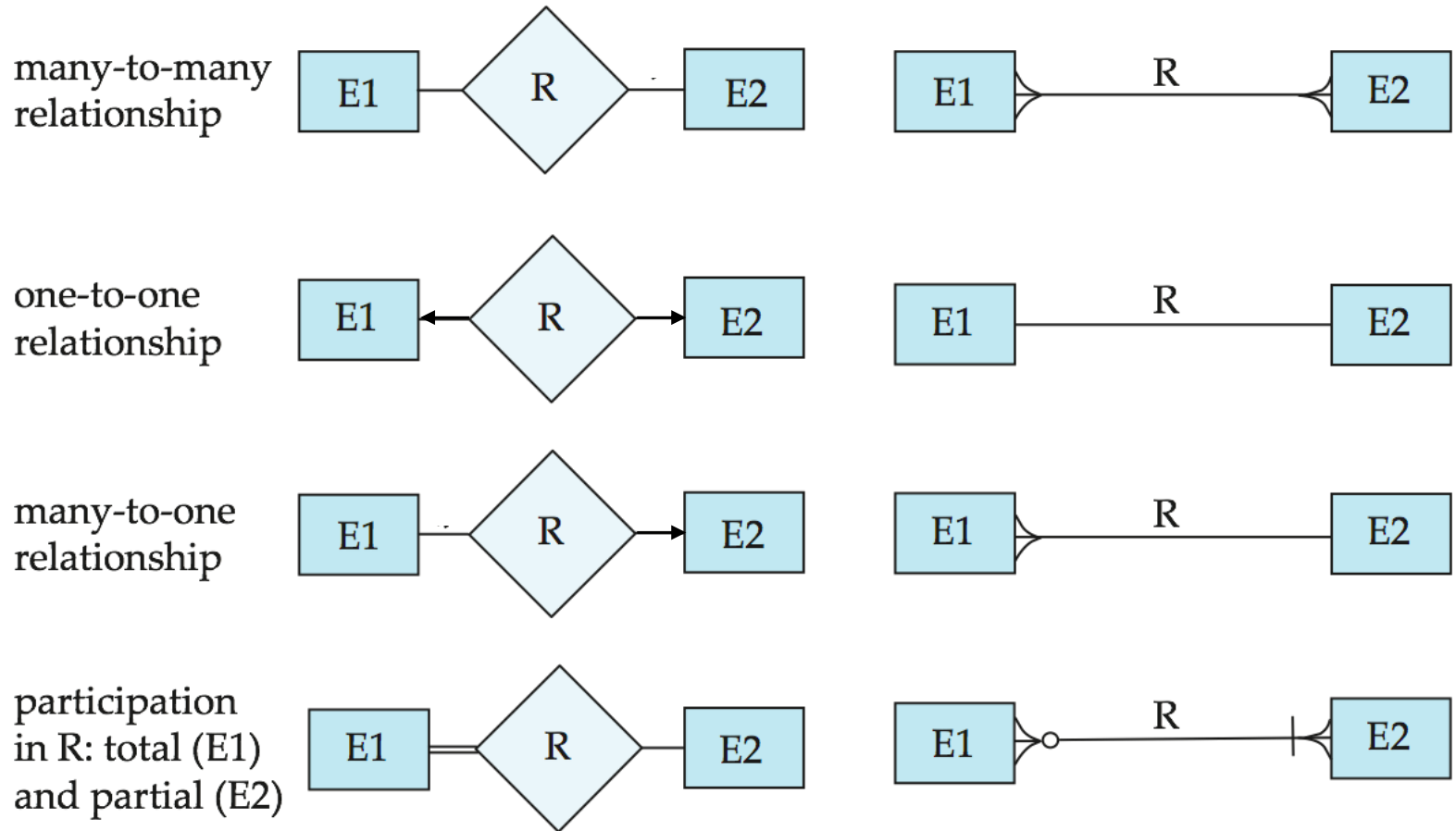


Equivalent in UML



Note reversal of position in cardinality constraint depiction

Other Diagram Notations: ER vs. Crows Feet Notation



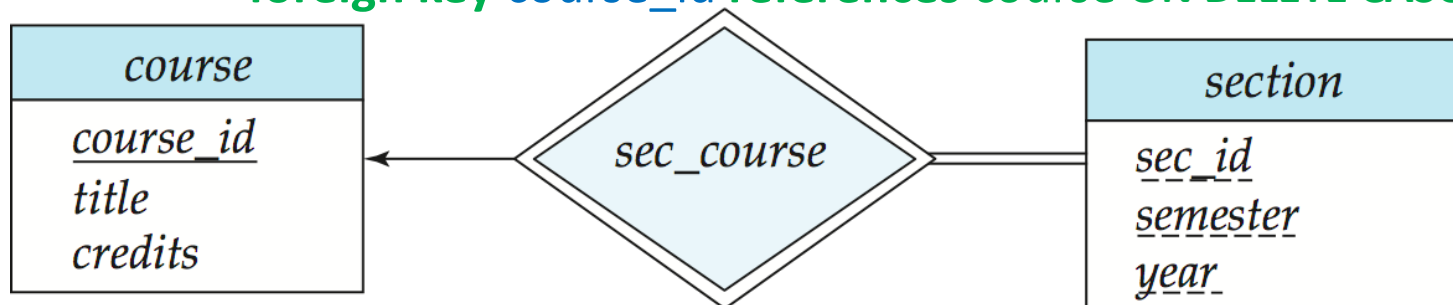
Converting E-R Diagrams to Relation Schemas

- Convert entity sets to relation schemas
 - special treatment of complex attributes
- Convert relationship sets to relation schemas

Convert Entity Sets to Relation Schemas

- First convert strong entity sets, then weak entity sets. If a weak entity set E_1 has another weak entity set E_2 as identifying entity set, then E_2 should be converted before E_1 .
- A strong entity set (with atomic attributes)
 - Becomes a relation schema with the same attributes and primary key
 - Example: *course(course_id, title, credits)*
- A weak entity set (with atomic attributes)
 - Becomes a relation schema that additionally includes the primary key of the identifying entity set as a foreign key attribute.
 - Example: *section(course_id, sec_id, semester, year)*

foreign key course_id references course ON DELETE CASCADE



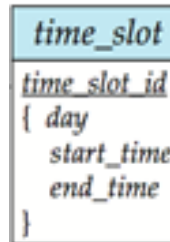
Conversion of Complex Attributes

- Composite attributes
 - Example: *name*, *address*
 - Fix: Store leaf attributes like *street_name* only
- Multivalued attributes A
 - Example: { *phone_number* }, a set of phone numbers.
 - Fix: Store it instead in a separate relation together with the primary key:
phones(*ID*, *phone_number*)
 foreign key *ID* references *instructor*
- Derived attributes
 - Example: *age()*, which can be computed from *date_of_birth*.
 - Fix: Store only *date_of_birth*, not *age* – it can be calculated from *date_of_birth* when needed.
- Conversion of the *instructor* entity set results hence in:
instructor(*ID*, *first_name*, ..., *street_number*, ..., *zip*, *date_of_birth*),
phones(*ID*, *phone_number*)

<i>instructor</i>
<u><i>ID</i></u>
<i>name</i>
<i>first_name</i>
<i>middle_initial</i>
<i>last_name</i>
<i>address</i>
<i>street</i>
<i>street_number</i>
<i>street_name</i>
<i>apt_number</i>
<i>city</i>
<i>state</i>
<i>zip</i>
{ <i>phone_number</i> }
<i>date_of_birth</i>
<i>age</i> ()

Clean Up: Remove Unnecessary Schemas

Example:



- Conversion of entity set results in:
 - *time_slot1(time_slot_id)*
 - *time_slot2(time_slot_id, day, start_time, end_time)*
- This is first simplified to:
 - *time_slot(time_slot_id, day, start_time, end_time)*
- And then to:
 - *time_slot(time_slot_id, day, start_time, end_time)*
as a timeslot with the same day and start time can't have different end times

Always remember not to follow conversion rules blindly – remember to clean up.

Convert Binary Relationship Sets to Relation Schemas

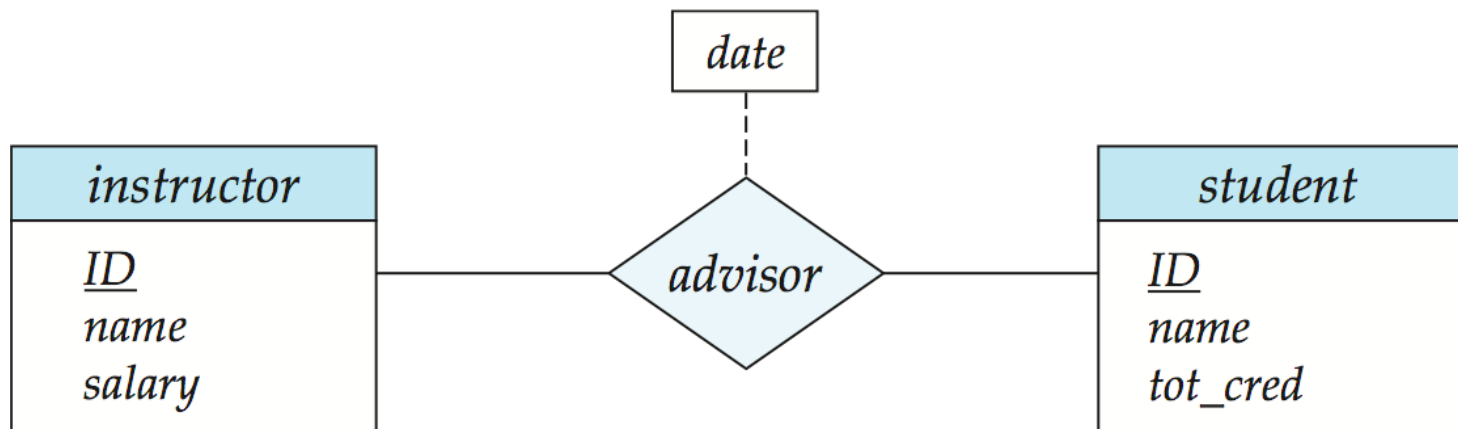
The conversion depends on the cardinality of the relationship set:

- Many-to-Many
- Many-to-One
- One-to-One

Conversion only needed for relationship sets which are not identifying relationship sets for a weak entity set.

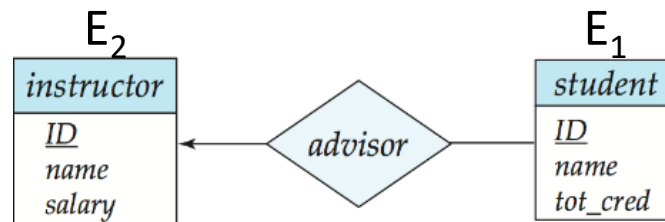
Convert Relationship Sets to Relation Schemas

- Many-to-Many relationship set R between E_1 and E_2
 - Is converted to a relation schema R with **attributes** from the primary keys of E_1 and E_2 and with any attributes of the relationship set.
 - The **primary key** of R is the union of the primary keys of E_1 and E_2 (and in very, very rare cases some of the relationship attributes of R , cf. slide 12).
 - The primary keys of E_1 and E_2 become **foreign keys** in R referencing E_1 and E_2 , respectively.
 - Example: Relation Schema for the relationship *advisor*
advisor(*student.ID*, *instructor.ID*, *date*)



Convert Relationship Sets to Relation Schemas

- Many-to-One relationship sets R between E_1 and E_2
 - **Approach 1:** Can be converted to a relation schema R in the same way as for Many-to-Many relationship sets R , but the primary key of the resulting relation schema is different: it is the primary key of E_1 .
 - **Approach 2:** It is possible to avoid making the new schema R , by instead modifying the schema of E_1 . This is explained on the following page. (For simplicity, it is assumed that the primary key of relationship set R does not include relationship attributes.)
 - **Approach 2 is usually preferred** unless E_1 has partial participation with only few entities in E_1 taking part in R , as in:

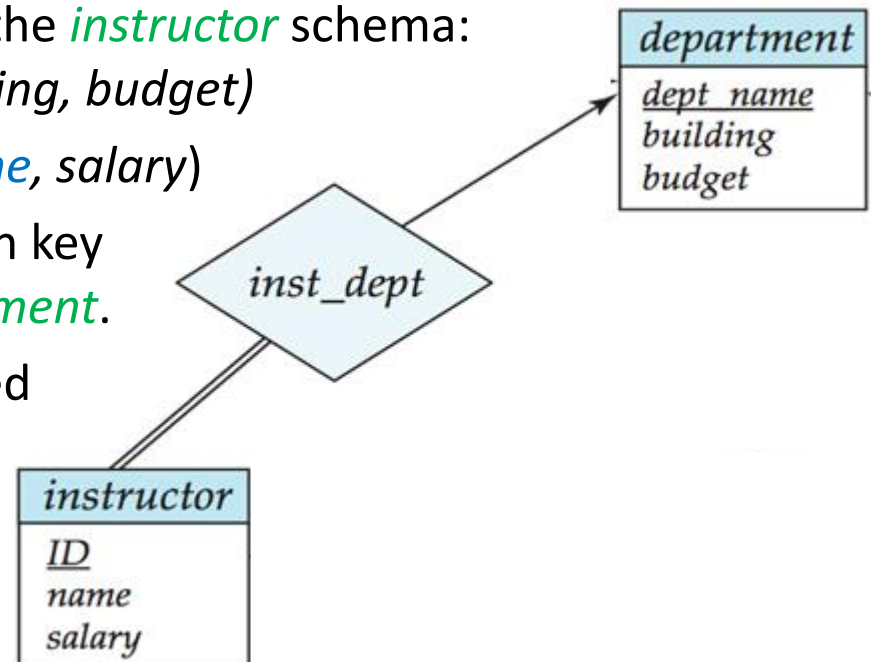


- One-to-One relationship sets R between E_1 and E_2
 - Similar comment. Approach 2 is preferred unless both E_1 and E_2 have partial participation with only few entities taking part in R .

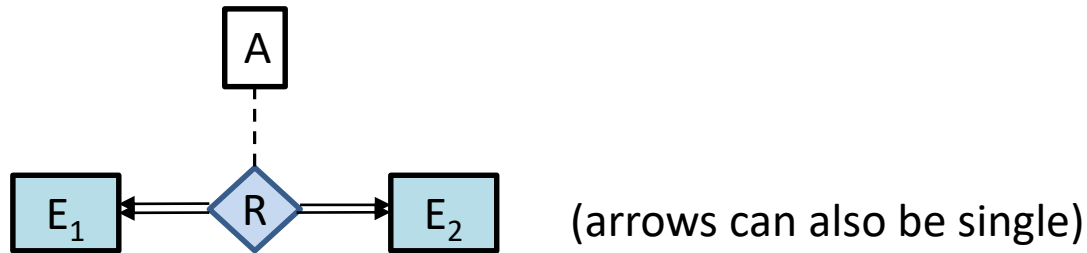
Convert Relationship Sets to Relation Schemas

■ Many-to-One relationship sets R from E_1 to E_2

- Add the attributes A of R and primary key K_2 of the E_2 (one-side) schema to the E_1 (many-side) schema. Hence, K_2 becomes a foreign key in E_1 :
 - $E_1(\underline{K_1}, \dots, A, K_2), E_2(\underline{K_2}, \dots)$
- Example: Instead of creating a schema for the relationship *inst_dept*, add an attribute *dept_name* to the *instructor* schema:
department(*dept_name*, *building*, *budget*)
instructor(*ID*, *name*, *dept_name*, *salary*)
- *instructor.dept_name* is a foreign key of *instructor* referencing *department*.
- Total participation can be defined by a "Not Null" constraint for *instructor.dept_name*.



Convert Relationship Sets to Relation Schemas



- **One-to-One relationship sets R between E_1 and E_2**
 - The primary key from one of the entity schemas can together with the attributes A of R be added to the other entity schema as a foreign key. This gives two possible solutions:
 - $E_1 (\underline{K}_1, \dots, A, K_2), E_2 (\underline{K}_2, \dots)$
 - $E_1 (\underline{K}_1, \dots), E_2 (\underline{K}_2, \dots, A, K_1)$
 - Choose the solution, which gives the least number of Null Values:
 - For solution 1 it holds that: only if participation of E_1 is partial, then for some rows in E_1 , K_2 will be Null.
 - For solution 2 it holds that: only if participation of E_2 is partial, then for some rows in E_2 , K_1 will be Null.
 - So if e.g. the participation of E_1 is total and E_2 is partial, then go for solution 1.



Summary

- E-R Modelling with E-R Diagrams
- Summary of Textbook Adapted UML Notation
- Converting E-R Diagrams to Relation Schemas
- Other diagram notations

Readings

- In Database Systems Concepts: see DTU Learn

Demo Exercises



Demo Exercises clarify ideas and concepts from the Lecture to provide you with good Database Skills.

Do the Demo Exercises. Solutions are found on later slides.

Super Key, Candidate Key & Primary Key

6.1 The Key concept

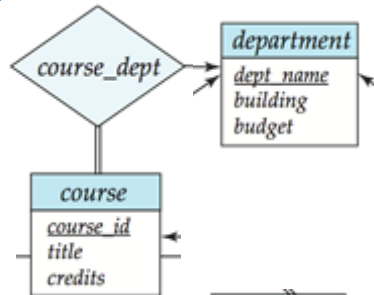
Explain the terms *super key*, *candidate key* and *primary key* for entities and relationships.

Entity-Relationship Diagram

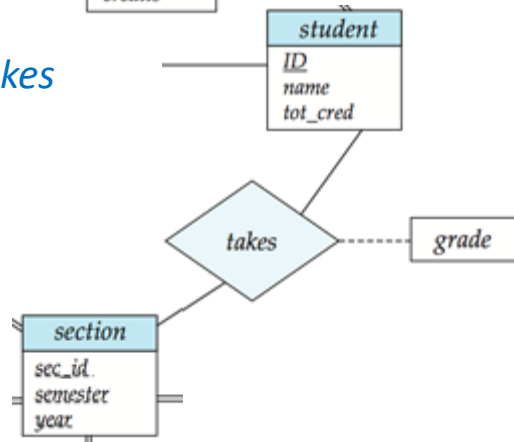
6.2 Reading an E-R Diagram

For the University E-R Diagram

a) Explain
course_dept

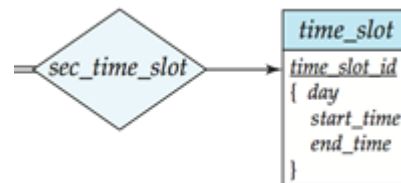


b) Explain *takes*



c)

Explain why the *time_slot* attribute {*day*,
start_time, *end_time*} is multivalued.



E-R Diagrams and Relation Schemas

6.3 From E-R Diagrams to Relation Schemas

An entity set E1 has the attributes A (primary key), B and C. Another entity set E2 has the attributes D (primary key), E and F. A relationship set R with cardinality one-to-many exists between E1 and E2. Both E1 and E2 have partial participation in R.

a) Draw an E-R Diagram

b) Convert the E-R Diagram to Relation Schemas

c) R is now given an attribute G.

Show the updated E-R Diagram and the updated Relation Schemas.

Diagram, Schemas and Instance

6.4 Draw E-R Diagram, Write Relation Schemas, Compose Database Instance

The manager of the Car Repair Shop stated that he has cars registered by license number, owner and model, on which he can provide various services registered by service number, type and price per hour. He tracks all repairs by license number, service number, delivery date and hours spent. The same car can have the same service several times (at different delivery dates).

- a) Construct an Entity-Relationship Diagram and specify the cardinality of the repairs relationship set.
- b) Convert the diagram to Relation Schemas and state any foreign keys.
- c) Construct a Database Instance.

Solutions to Demo Exercises



Super Key, Candidate Key & Primary Key

6.1 The Key concept

Explain the terms *super key*, *candidate key* and *primary key* for entities and relationships.

A **super key** is a set of one or more attributes that, taken collectively, identifies uniquely an entity in the entity set, or a relationship in a relationship set. A super key may contain extraneous attributes. If K is a super key, then so is any superset of K.

A **candidate key** is a super key, for which no proper subset exists (i.e. a minimal super key). It is possible that several distinct sets of attributes could serve as candidate keys.

A **primary key** is one of the candidate keys that is chosen by the database designer as the principal means of identifying elements in an entity set or associations in a relationship set.

Entity-Relationship Diagram

6.2 Reading an E-R Diagram

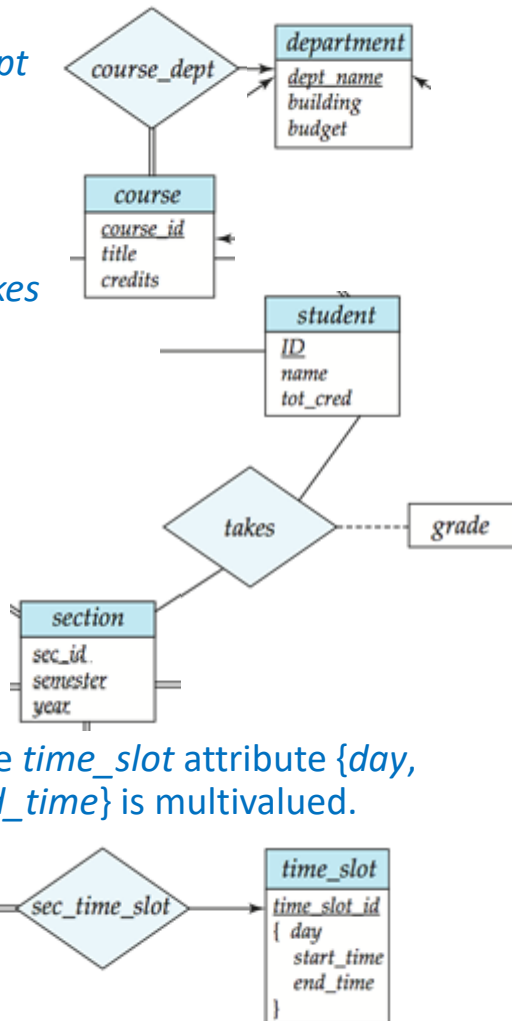
For the University E-R Diagram

a) Explain *course_dept*

b) Explain *takes*

c)

Explain why the *time_slot* attribute {*day*, *start_time*, *end_time*} is multivalued.



a) *course_dept* defines relationships between courses and departments. It has a **many-to-one** cardinality (shown by an arrowhead towards department), and the participation of course is **total** (is shown by a double line), while the participation of department is **partial** (is shown by a single line). This together means that a *course* is always offered by exactly one *department*, and a *department* can offer from zero to several *courses*.

b) *takes* defines relationships between students and sections. It has a **many-to-many** cardinality (as there are no arrowheads), and the participation is **partial** on both sides. This together means that a *student* may take from zero to several *sections*, and that a *section* may be taken by zero to several students. *takes* has **relationship attribute** *grade* shown by a dotted line.

c) A *time_slot_id* should define a set of weekly meetings (each given by the day, start and end time), for instance Monday, 08.00-10.00 and Thursday, 13.00-15.00. As we need *a set* of meeting times, the attribute is multivalued.

E-R Diagrams and Relation Schemas

6.3 From E-R Diagrams to Relation Schemas

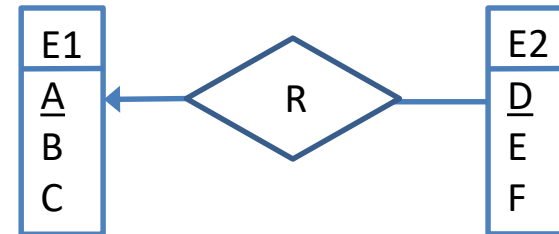
An entity set E1 has the attributes A (primary key), B and C. Another entity set E2 has the attributes D (primary key), E and F. A relationship set R with cardinality one-to-many exists between E1 and E2. Both E1 and E2 have partial participation in R.

a) Draw an E-R Diagram

b) Convert the E-R Diagram to Relation Schemas

c) R is now given an attribute G.
Show the updated E-R Diagram and the updated Relation Schemas.

E-R Diagram



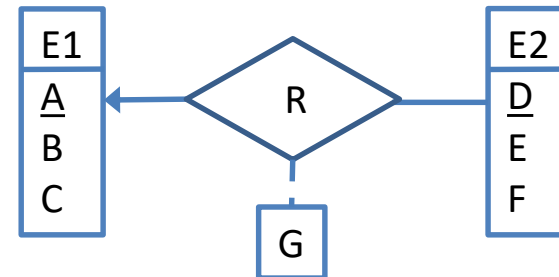
Relation Schemas

R1(A, B, C)

R2(D, E, F, A)

Relationship R is stored in R2 i.e. the 'many' side by adding A as a foreign key.

Updated E-R Diagram



Updated Relation Schemas

R1(A, B, C)

R2(D, E, F, A, G)

The R attribute(s) G are stored in R2.

Diagram, Schemas and Instance

6.4 Draw E-R Diagram, Write Relation Schemas, Compose Database Instance

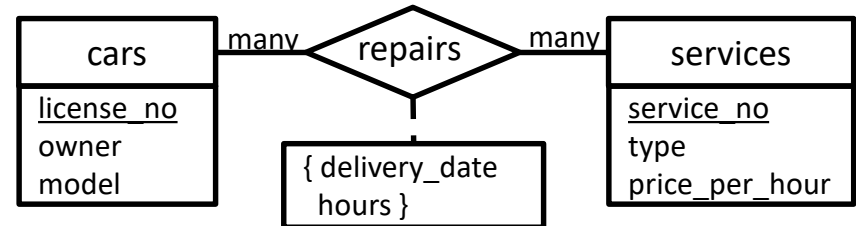
The manager of the Car Repair Shop stated that he has cars registered by license number, owner and model, on which he can provide various services registered by service number, type and price per hour. He tracks all repairs by license number, service number, delivery date and hours spent. The same car can have the same service several times (at different delivery dates).

a) Construct an Entity-Relationship Diagram and specify the cardinality of the repairs relationship set.

b) Convert the diagram to Relation Schemas and state any foreign keys.

c) Construct a Database Instance.

Entity-Relationship Diagram



Relation Schemas

cars(license_no, owner, model)

services(service_no, type, price_per_hour)

repairs(license_no, service_no, delivery_date, hours)

repairs has foreign keys license_no referencing cars and service_no referencing services.

Database Instance

license_no	owner	model
ZY43816	Adam Asimov	Honda Accord 2,0 Aut
YE26036	Brian Balter	Audi A4 Limousine 2.0
UZ58368	Brian Balter	Alfa Romeo 159 2,1

service_no	type	price_per_hour
431001	Bodywork	315,50
431002	Painting	345,50
451001	Parts replacement	216,50
451002	Parts repair	389,50

license_no	service_no	delivery_date	hours
ZY43816	431001	2015-02-17	2,5
ZY43816	431002	2015-02-17	1,5
UZ58368	51001	2015-02-18	0,5

Exercises

Please answer all exercises
to demonstrate your
Database Skills.

Pencil and Paper Exercises
Solutions are available at 11:45



Diagrams, Schemas and Instance

6.5 Draw Diagram and Write Schemas

An entity set E1 has the attributes A (primary key), B and C. Another entity set E2 has the attributes D (primary key), E and F.

A relationship set R with cardinality many-to-many exists between E1 and E2. Both E1 and E2 have partial participation in R.

- a) Draw an E-R Diagram.
- b) Convert the E-R Diagram to Relation Schemas and state any foreign key constraints.
- c) The relationship is given an attribute G. Show the updated E-R Diagram and the updated Relation Schemas.

6.6 Sets, Diagram, Schemas and Instance

A car insurance company wants a database to

- (1) track clients with name, address and age,
- (2) track cars, with license plate number, model, color and production year, and
- (3) track the car ownerships of clients, each with a start date. It is assumed that a car is owned by exactly one client.

- a) List Entity Sets and Relationship Sets.
- b) Make an E-R Diagram of the Car Insurance Company.
- c) Convert the E-R Diagram to Relation Schemas and state any foreign key constraints.
- d) Make an example of a Database Instance.

Diagrams, Schemas and Instance

6.7 Draw Diagram and Write Schemas

A company needs help keeping track of the transactions through some bank's ATMs. They have requested a simple database to help with this. An ATM has an ID, a Location and an Associated Bank. A transaction has a transaction number and an amount and is identified by the ID from ATM and the transaction number.

- a) Make an E-R Diagram describing the company.
- b) Convert the E-R Diagram to Relation Schemas and state any foreign key constraints.
- c) Draw the associated database schema diagram.